



Before using this Quick Reference, read and observe the detailed Rolls-Royce Solutions overall documentation on BlueVision!
In particular, all safety instructions in the overall documentation must be observed!

Third-party remote control system

Connecting a third-party remote control system (shipyard's supply) to LOP and GCU

Points of connection

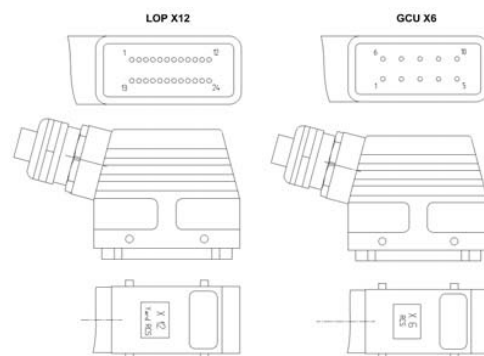
The third-party remote control system must be connected to the connectors X12 on the LOP and to X6 on the GCU.

Functions

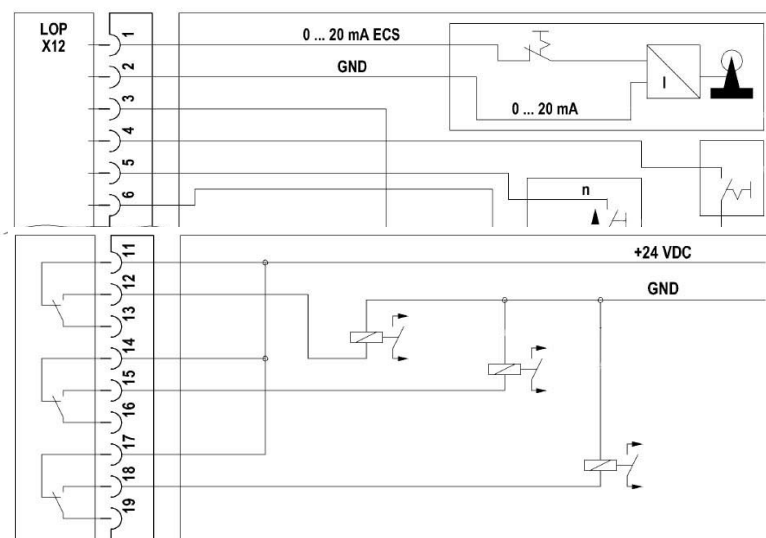
Inputs and outputs are provided for the following functions:

- Analog engine speed setting signal (4 mA to 20 mA), speed setting via binary signals "SPEED INCREASE / SPEED DECREASE", selection of speed setting mode: Binary or analog.
- Clutch command signals (binary inputs).
- Disengagement command from local control / monitoring system to remote control system, idling speed and local mode status.
- Engine speed in the speed range where the clutch can be operated ("dis/engagement window").
- Clutch control feedback signal.

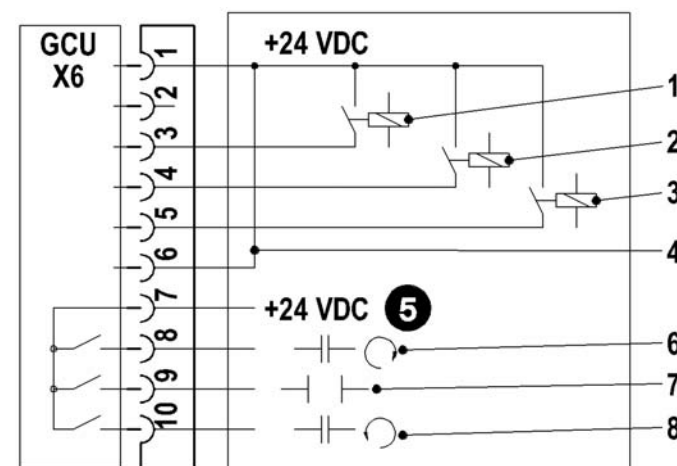
Connectors LOP X12 and GCU X6



Ensure correct position of inserts when assembling the connector housings.



Pin No.	Signal	Remarks
1	Input of analog speed setting signal	4 to 20 mA
2	Ground	
3	Output voltage	Power supply +24 VDC ECS
4	Selection of speed setting: Binary or analog.	Optocoupler input (+24 VDC). Must be connected to Pin 3.
5	Binary signal "Speed increase".	
6	Binary signal "Speed decrease".	
7	Analog voltage signal 0 to 10 VDC.	Proportional to engine speed.
8	GND	Ground analog
9	Analog voltage signal 0 to 10 VDC.	Engine load signal / fuel injection not used
10	GND	Ground analog
11	Local mode active	Common
12	(potential-free relay output)	NO contact
13	max. 0.5 A.	NC contact
14	Disengagement command and/or idling speed setting signal (potential-free relay output)	Common
15	at start / stop / emergency stop (max. 0.5 A).	NO contact
16		NC contact
17	Dis/engagement window	Common
18	(potential-free relay output) max. 0.5 A.	NO contact
19		NC contact



- 1 Command signal PARALLEL ROTATION / FLUSH
- 2 Command signal NEUTRAL / DISENGAGEMENT RELEASE / DISENGAGE
- 3 Command signal COUNTER-ROTATION / ENGAGE
- 4 Third-party remote control system connected
- 5 U_{ext} from third-party remote control system
- 6 Feedback signal COUNTER-ROTATION / (ENGAGED)
- 7 Feedback signal NEUTRAL / (DISENGAGED)
- 8 Feedback signal COUNTER-ROTATION / FLUSH

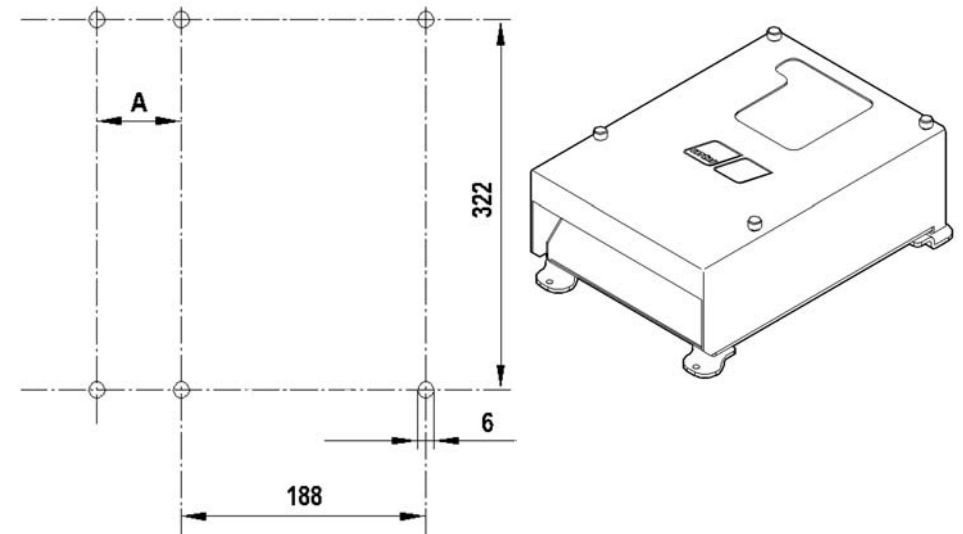
Pin No.	Signal	Remarks
1	Third-party RCS, third-party RCS is connected.	Signal +24 VDC from GCU. Must be connected to input of Third-party RCS (Pin 6).
2	GND	Ground
3	Command signal PARALLEL ROTATION 1 / FLUSH 2	Must be connected to +24 VDC X6-1 (max. 2.0 A).
4	Command signal NEUTRAL 1 / DISENGAGE 2, 3	
5	Command signal COUNTER-ROTATION 1 / ENGAGE 2, 3	
6	Third-party RCS, third-party RCS is connected.	$U_{ext} = +24 \text{ VDC}$
7	Input voltage U_{ext}	+ 24 VDC, max. 0.2 A
8	Feedback signal PARALLEL ROTATION 1 / ENGAGED 2, 3	+ 24 VDC, max. 0.2 A
9	Feedback signal NEUTRAL 1 / DISENGAGED 2, 3	+ 24 VDC, max. 0.2 A
10	Feedback signal COUNTER-ROTATION 1 / FLUSH 2	+ 24 VDC, max. 0.2 A

- 1 Gearbox with reversing gear, FPP
- 2 Gearbox with reversing gear, WJ (with flush feature)
- 3 Gearbox w/o reversing gear, WJ / CPP

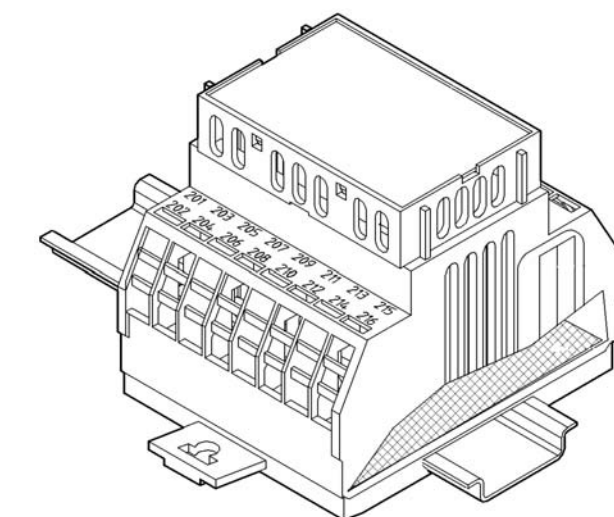
Third-party monitoring system

Mechanical installation

Installing PIM 4 module housing



Installing PIM 5 for VDR connection



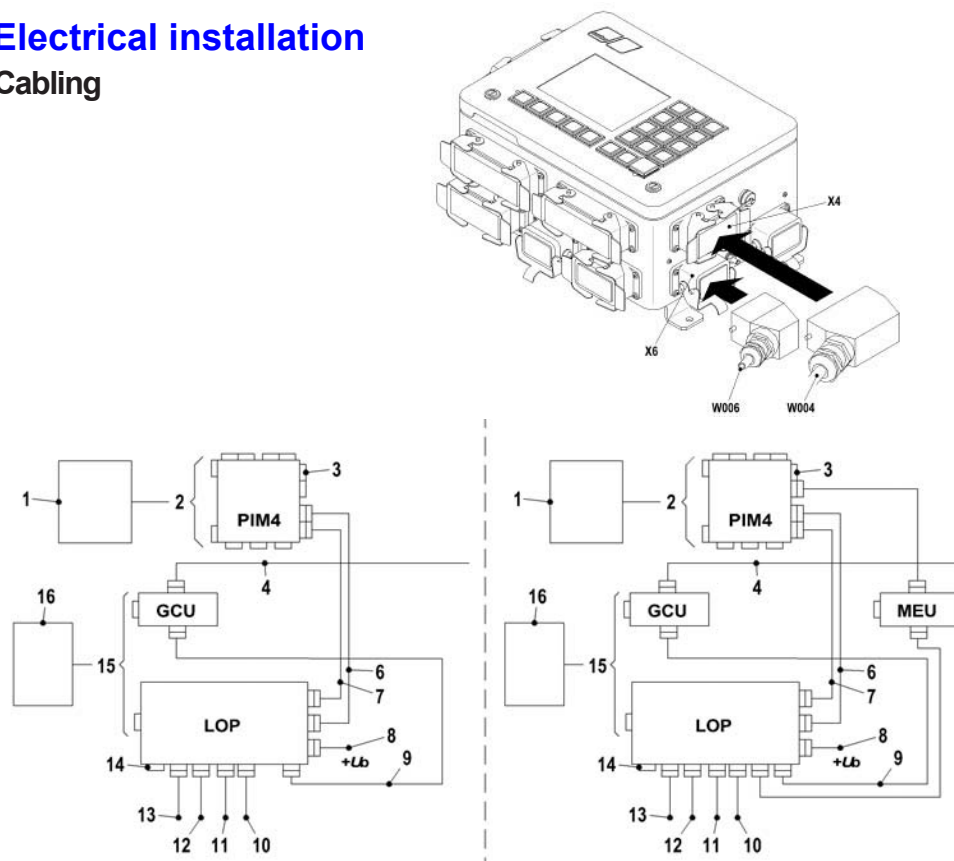
Module PIM 5 is mounted on top-hat rails.



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Electrical installation

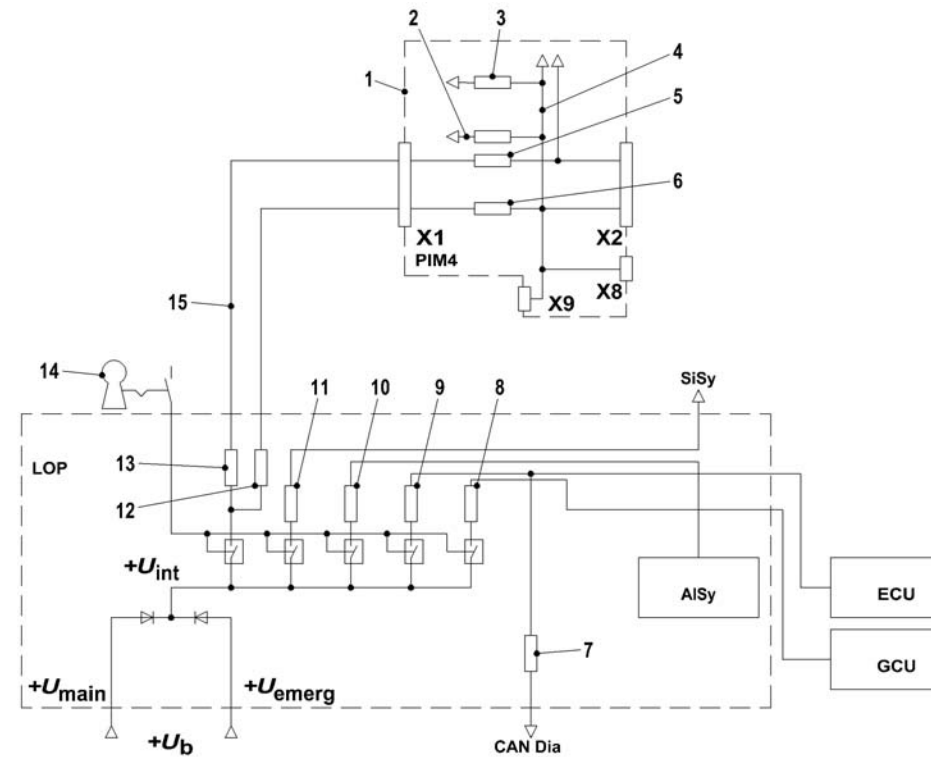
Cabling



No.	Name	Explanation	Type	Supplier
1	-	Third-party monitoring system	-	Shipyard
2	Cable	Interface to Third-party monitoring system	Dat, Sig	Shipyard
3	-	PIM 4 control unit	-	-
4		Connection to monitoring system extension MEU in systems with extended scope of measuring points	Sig	Rolls-Royce Solutions
5		Connection between Gear Control Unit GCU and gearbox	Sig	Rolls-Royce Solutions
6	Cable	Connection between X4 at control stand LOP and X3 at PIM 4 control unit	Dat, Sig	Rolls-Royce Solutions
7	Cable	Connection between X6 at control stand LOP and X1 at PIM 4 control unit	Spg	Rolls-Royce Solutions
8	Cable	Overall power supply	Spg	Rolls-Royce Solutions
9	Cable	Connection between control stand LOP and Gear Control Unit GCU	Spg	Rolls-Royce Solutions
10	Cable	Connection between control stand LOP and engine governor ECU	Spg	Rolls-Royce Solutions
11	Cable	Connection between control stand LOP and engine governor ECU	Sig	Rolls-Royce Solutions
12	Cable	Connection between control stand LOP and engine sensor"water in fuel prefilter" 1 (option: 2 cables for 2nd sensor) and facility to connect a horn and/or a beacon in engine room.	Sig	Rolls-Royce Solutions

No.	Name	Explanation	Type	Supplier
13	Cable	Connection between control stand LOP and terminal box X010.	Sig	Rolls-Royce Solutions
14	Connector	To connect additional sensors in the engine room (shipyard's supply)	Sig	Rolls-Royce Solutions
15	Cable	Interface to third-party remote control system	Sig	Shipyard
16	-	Third-party remote control system	-	Shipyard

Power supply



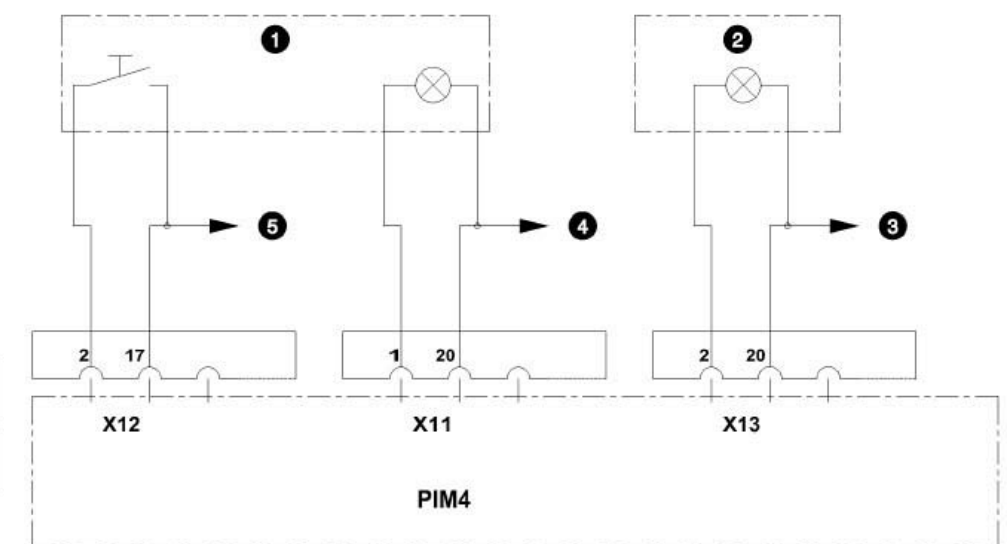
No.	Function name	Explanation
1		PIM 4 control unit
2	F1 on BOB 1 (1) - 3.15 A	Supply for indicator lamps of the control pushbuttons on the PIM 4 control unit.
3	F1 on BOB 1 (2) - 3.15 A	Supply of alarm indicator lamps on the PIM 4 control unit.
4	+U_bint	Internal supply voltage.
5	F16 - 15 A	Input for PIM 4 control unit internal supply voltage .
6	F19 - 10 A	Input for internal supply voltage.
7	F1 - 4 A	Supply for CAN dialog unit
8	F69 - 4 A	Supply for gear control unit GCU.
9	F67 - 20 A	Supply for ECU and externally connected dialog units.
10	F70 - 15 A	Supply voltage for ALSY alarm system.
11	F71 - 3 A	Supply voltage for safety system.
12	F2 - 15 A	Output circuit for control stands' power supply (located at the located on printed circuit board PSB, in LOP).
13	F1 - 15 A	Output circuit for control stands' power supply (located at the located on printed circuit board PSB, in LOP).
14	S001	Key switch
15	+UOut	Supply voltage for PIM 4 control unit.

The starter is directly connected to the starter battery and supplied from there. The supply voltage +Ub for all mtu devices is applied to the LOP of the ECS.

Connecting a third-party monitoring system to the PIM 4 control unit

The indicator lamps, pushbuttons, pushbutton lamps and the display instruments are to be connected to the connectors X10, X11, X12, X13, X14 (included in the scope of supply) and (if fitted) to a MCS at connector X18. It is recommended to split these devices into the following groups and to connect each of these groups with a separate cable to the corresponding connector

- Group 1: Parallel interface; control pushbuttons, key switch, horn, pushbutton lamps, indicator lamps for preheating (if applicable) "engine running" and "ready for operation" connectors X10, X11, X12 and X18 (for connection of MEU in systems with extended scope of measuring points).
- Group 2: Parallel interface; all alarm indicator lamps, connector X13.
- Group 3: Parallel interface; all display instruments and their illumination, connector X14.
- Group 4: Serial interface, connector X17.



- 1 Start
- 2 OVERSPEED
- 3 to further indicator lamps
- 4 to further lamps
- 5 to further control pushbuttons

Pin assignment

